Data Science

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What is data science?

- Data science is using data to answer questions. This is a pretty broad definition, and that's because it's a pretty broad field!
- Data science can involve:
  - Statistics, computer science, mathematics
  - Data cleaning and formatting
  - Data visualization

An Economist Special Report sums up this melange of skills well - they state that a data scientist is broadly defined as someone:

"who combines the skills of software programmer, statistician and storyteller slash artist to extract the nuggets of gold hidden under mountains of data"

And by the end of these courses, hopefully you will feel equipped to do just that!

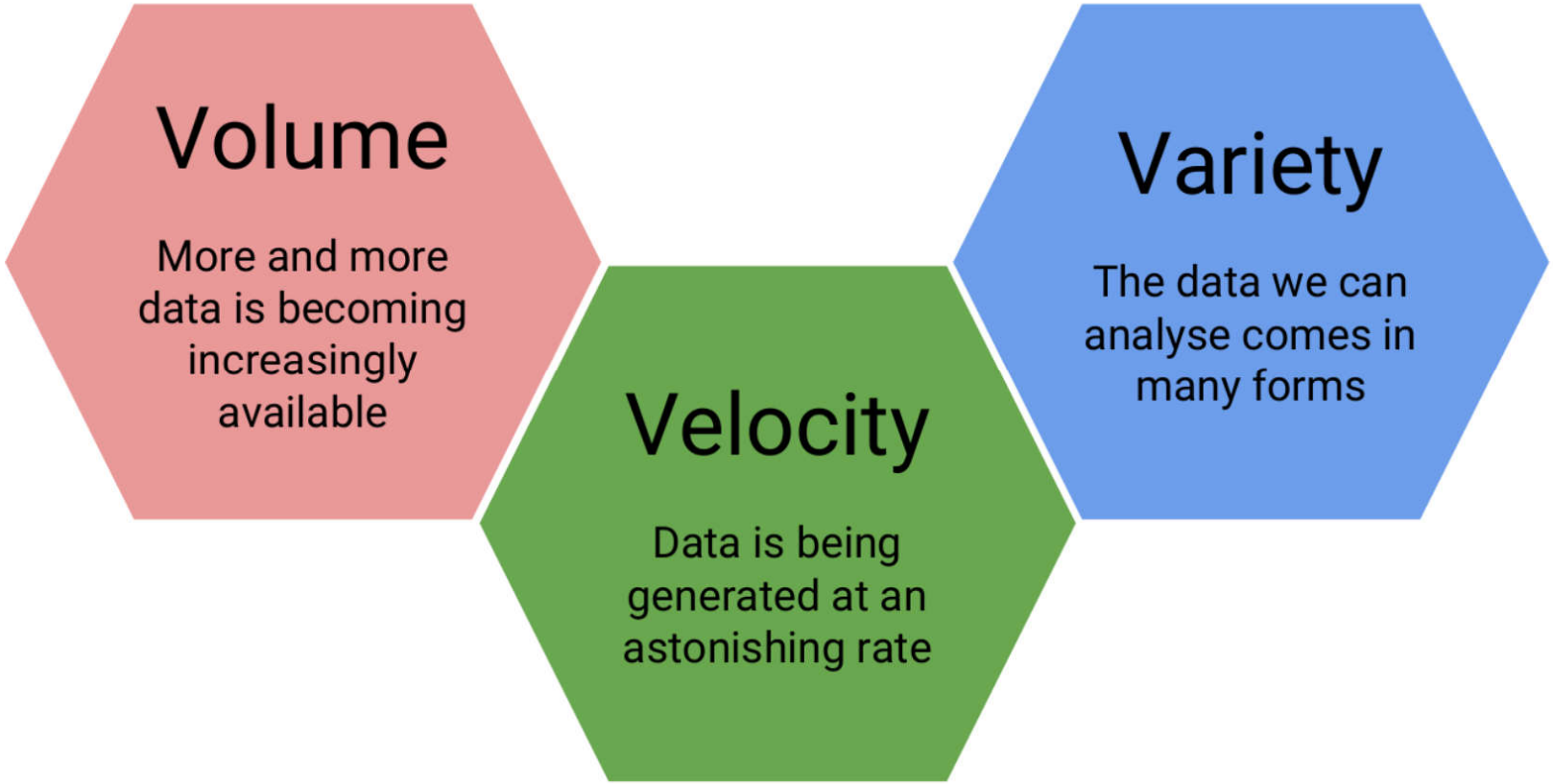
Why do we need data science?

- One of the reasons for the rise of data science in recent years is the **vast amount of data** currently available and being generated.

- Not only are massive amounts of data being collected about many aspects of the world and our lives, but we simultaneously have the **rise of inexpensive computing**.

- This has created the perfect storm in which **we have rich data and the tools to analyze it**: Rising computer memory capabilities, better processors, more software and now, more data scientists with the skills to put this to use and answer questions using this data!

Big Data


Volume
More and more data is becoming increasingly available
Velocity
Data is being generated at an astonishing rate
Variety
The data we can analyse comes in many forms

Big Data

IBM's Four Vs of big data, which differs from the other definitions by adding a fourth V, *veracity*, into the mix:

Volume: The first V is volume (measured in gigabytes or petabytes of data) based on the amount of data being produced and the amount being captured.
Variety: The second V is variety. Variety refers to the number of different forms of big data being captured. It can be a combination of social media data and structured or infrastructure data including sensory data (coming from environmental sensors), enterprise data, or transactional data.
Velocity: The third V is velocity. Velocity refers to how quickly the data is moving (the data in motion).
Veracity: The fourth V is veracity, which is associated with uncertainty. It relates to how you can make decisions when you don't know how accurate the data is (i.e., is the data good enough to support a particularly important question)?

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Skills of Successful Data Scientists

Business and domain expertise: This involves articulating how information, insights, and analytics can help business leaders answer key questions—and determining *which* questions need answering—to make appropriate decisions. Data scientists need a thorough understanding of their business across the value chain (marketing, sales, distribution, operations, pricing, products, finance, risk, etc.) to be successful. We often pair data scientists (who are experts in analytics) with subject matter experts (who are knowledgeable about their type of business) to create an even stronger fusion of skills.

Skills of Successful Data Scientists

Expertise in machine learning algorithms: This is the ability to identify the most appropriate analytical techniques for addressing different classes of problems, applying relevant techniques, translating the results, and generating insights that provide valuable business outcomes. To do this, a data scientist must first have a thorough understanding of machine learning (e.g., predictive modeling, cluster analysis, and optimization techniques) as well as techniques, tools, and languages used to run analyses (such as Python, R, IBM SPSS Modeler, SAS Enterprise Miner, or KNIME Analytics Platform).

Skills of Successful Data Scientists

Expertise in software engineering: This involves determining the appropriate software packages or modules to run, the ability to modify them, and the ability to design and develop new computational techniques to solve business problems (e.g., machine learning, natural language processing, graph/social network analysis, neural nets, and simulation modelling). Frequently, data scientists have some computer science expertise and are comfortable designing and deploying their solutions both on-prem as well as in the cloud.

Expertise in data storage (both traditional relational databases and noSQL databases) and data engineering: This is a thorough understanding of external and internal data as well as how they are gathered, stored, and retrieved. This will enable a data scientist—and by extension the business as a whole—to:

1. Extract, transform and load data stores.
2. Retrieve data from external sources (through screen scraping and data transfer protocols).
3. Use and manipulate large “big data” data stores (like Hadoop, Spark, MongoDB and Elasticsearch).
4. Use the disparate data sources to analyze the data and generate insights.

Skills of Successful Data Scientists

Visual presentation and communications expertise: This involves thoroughly understanding visual art and design, which is important because effective visual presentations enable business leaders (who are not professional data analysts) to interpret data and embrace it in their decision making processes. Accordingly, a data scientist should be able to

1.Take statistical and computational analyses and turn them into graphs, charts, and animations.

2.Create visualizations (e.g. motion charts, word maps) that clearly show insights from data and corresponding analytics.

3.Generate static and dynamic visualizations using a variety of visual media including printed reports, digital display devices (screens on mobile devices, laptops, desktop computers, and large-format HD visualization “walls”), interactive applications, and—perhaps soon—augmented reality glasses.

Last, but not least, a data scientist should be able to engage with senior management, talk their language, and translate data-driven insights into decisions and actions.

Course Outlines

- Data Science lifecycle
- Types of data
- Statistical Inference
- Machine learning
  - Supervised learning (Regression/Classification)
    - Probabilistic learning
    - Generative models
    - Bayesian models
    - Neural networks
    - Evaluation metrics
    - Kernel methods
  - Unsupervised learning
    - Dimentionality Reduction
    - Clustering
- Deep learning
  - CNN-RNN-NLP-GNN


DATA SCIENCE LIFECYCLE
01
BUSINESS UNDERSTANDING
Ask relevant questions and define objectives for the problem that needs to be tackled.
02
DATA MINING
Gather and scrape the data necessary for the project.
03
DATA CLEANING
Fix the inconsistencies within the data and handle the missing values.
04
DATA EXPLORATION
Form hypotheses about your defined problem by visually analyzing the data.
05
FEATURE ENGINEERING
Select important features and construct more meaningful ones using the raw data that you have.
06
PREDICTIVE MODELING
Train machine learning models, evaluate their performance, and use them to make predictions.
07
DATA VISUALIZATION
Communicate the findings with key stakeholders using plots and interactive visualizations.

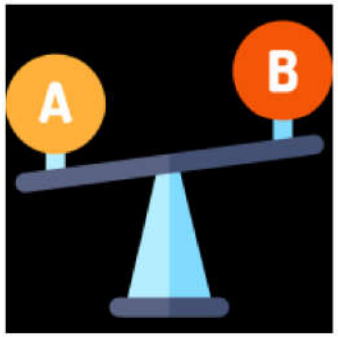
1. BUSINESS UNDERSTANDING

- A project starts by understanding the *what*, the *why*, and the *how* of your project.
- The outcome of this phase:
  - clear research goal
  - a good understanding of the context
  - well-defined deliverables
  - a plan of action with a timetable and cost estimate
- The design team should think carefully about the use scenario
  - The business problem will be mapped to data science tasks.

Problem Definition

- **Define objectives:** work with your customer to understand and identify the business problems.
- **Formulate questions:** convert the business goals into questions that the data science techniques can target.
- **Define the success metrics:** look for specific, measurable, achievable, relevant, and time-bound metrics.
- **Identify data sources:** look for the data that is relevant to the question.

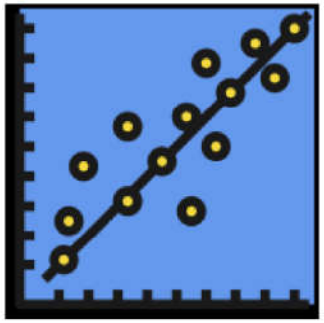
Formulate Questions



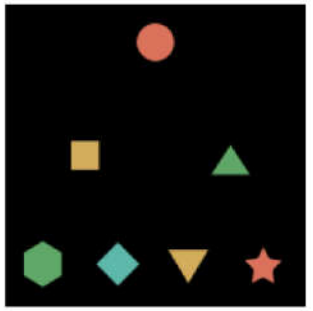
Comparison



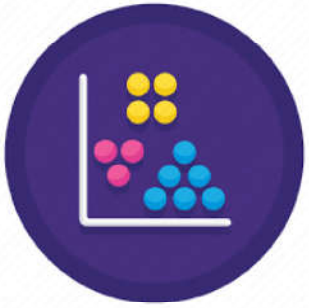
Description



Regression



Classification



Clustering



Anomaly Detection



Recommendation

Define Success Metric

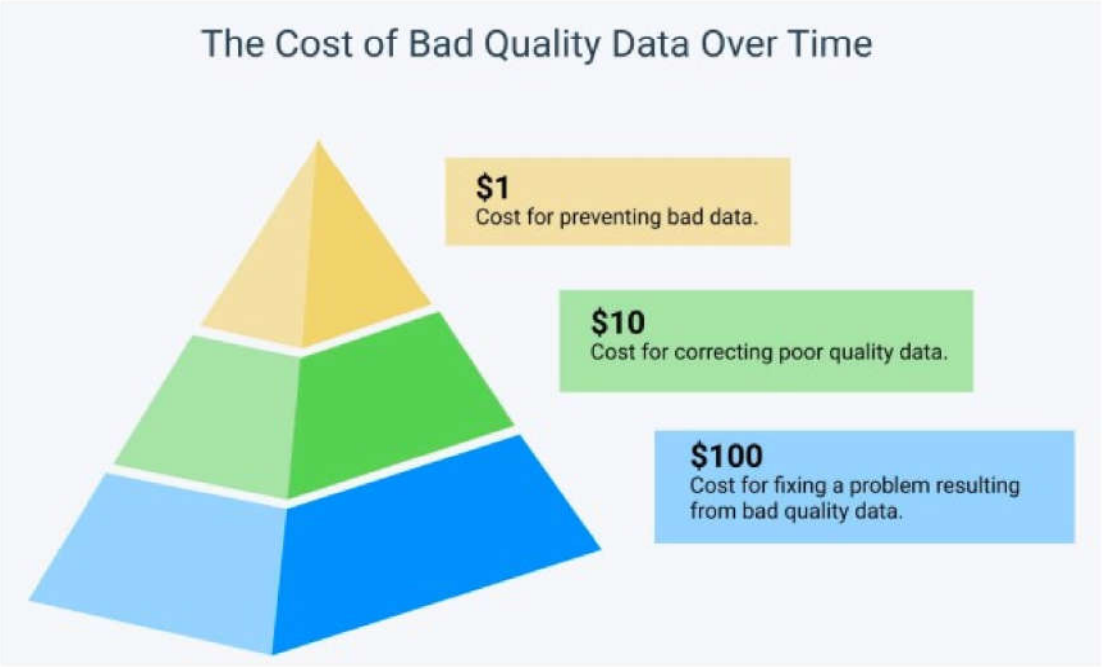
- Most companies don’t care about the fancy ML metrics.
- The sole purpose of businesses: maximize profits.
- In case of Netflix:
  - The objective is to increase revenue by 5%.
  - To increase revenue, we need to increase the customer retention rate by 8%.
  - To increase the customer retention rate, we need to increase the accuracy of the recommender system by 10%.
- Look for specific, measurable, achievable, relevant, and time-bound metrics.

Data Collection

- **Data collection** is the process of gathering and measuring information of interest, in an established systematic fashion that enables one to answer stated research questions, test hypotheses, and evaluate outcomes.
  - What data do I need for my project?
  - Where does it live?
  - How can I obtain it?
  - What is the most efficient way to store and access all of it?

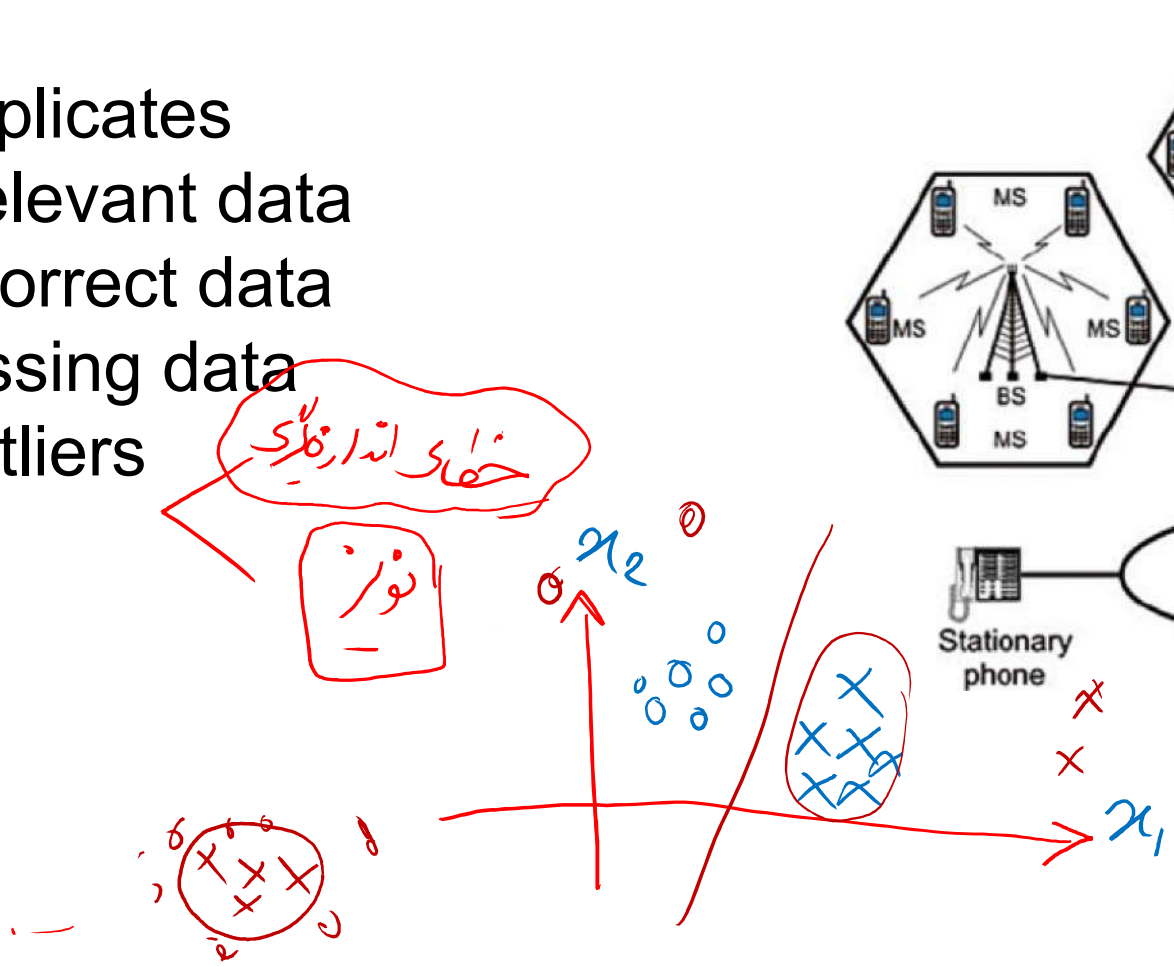
Data Cleaning

- Data cleaning is the process of editing, correcting, and structuring data within a data set so that it's generally uniform and prepared for analysis.



Data Cleaning

- Duplicates
- Irrelevant data
- Incorrect data
- Missing data
- Outliers



Standardize + Normalize

- Standardization and normalization make data ripe for statistical analysis and easy to compare and analyze.
- **Standardization** is a process during which you're making sure all your values adhere to a specific standard:
  - Deciding whether to go with kilos or grams, upper or lower case, etc.
  - Example: +989121234567, 00989121234567, 989121234567, 09121234567 → 9121234567
- **Normalization** is the process of adjusting the values to a common scale.
  - Example: rescale values into the 0-1 range.

Feature Engineering

- Feature engineering is the process of using domain knowledge to transform your raw data into informative features.
- This step requires a creative combination of domain expertise and the insights obtained from the data exploration step.
- This stage will directly influence the accuracy of the predictive model you construct in the next stage.

Feature Engineering

- Feature selection: is the process of cutting down the features that add more noise than information.
  - **Filter methods**: apply statistical measure to assign scoring to each feature
  - **Wrapper methods**: frame the selection of features as a search problem and use a heuristic to perform the search
  - **Embedded methods**: use machine learning to figure out which features contribute best to the accuracy
- Feature construction: involves creating new features from the ones that you already have.

Predictive Modeling



- Predictive modeling is where the machine learning finally comes into your data science project.
- Depending on the type of question that you're trying to answer, there are many modeling algorithms available.
- The models that you train will be dependent on:
  - the size, type and quality of your data
  - how much time and computational resources you are willing to invest
  - the type of output you intend to derive.

Data Visualization

- Data visualization combines the fields of communication, psychology, statistics, and art, with an ultimate goal of communicating the data in a simple yet effective and visually pleasing way.

- Present your solution:
  - Highlighting what you have learned
  - Expose the model with an interface
    - Data Dashboards



What is data?

First, let's look at what a few trusted sources consider data to be.

We'll look at the **Cambridge English Dictionary**, which states that data is:

Information, especially facts or numbers, collected to be examined and considered and used to help decision-making.

Second, we'll look at the definition provided by **Wikipedia**, which is:

A set of values of qualitative or quantitative variables.

What is data?

So, the first thing to focus on is **“a set of values”** - to have data, you need a set of **items** to measure from. In statistics, this set of items is often called the **population**. The set as a whole is what you are trying to discover something about. For example, that set of items required to answer your question might be **all websites** or it might be the set of **all people coming to websites**, or it might be a set of **all people getting a particular drug**. But in general, it’s a set of things that you’re going to make measurements on.

What is data?

The next thing to focus on is **“variables”** - variables are measurements or characteristics of an item. For example, you could be measuring the height of a person, or you are measuring the amount of time a person stays on a website. On the other hand, it might be a more qualitative characteristic you are trying to measure, like what a person clicks on a website, or whether you think the person visiting is male or female.
Finally, we have both **qualitative and quantitative** variables.

Qualitative variables are information about qualities. They are things like **country of origin, sex, or treatment group**. They’re usually described by words, **not numbers**, and they are not necessarily ordered.

What is data?

Quantitative variables on the other hand, are information about quantities. Quantitative measurements are usually described by **numbers** and are measured on a continuous, ordered scale; they're things like **height, weight and blood pressure**.

"A set of values of qualitative or quantitative variables"

Set: In statistics, the population you are trying to discover something about

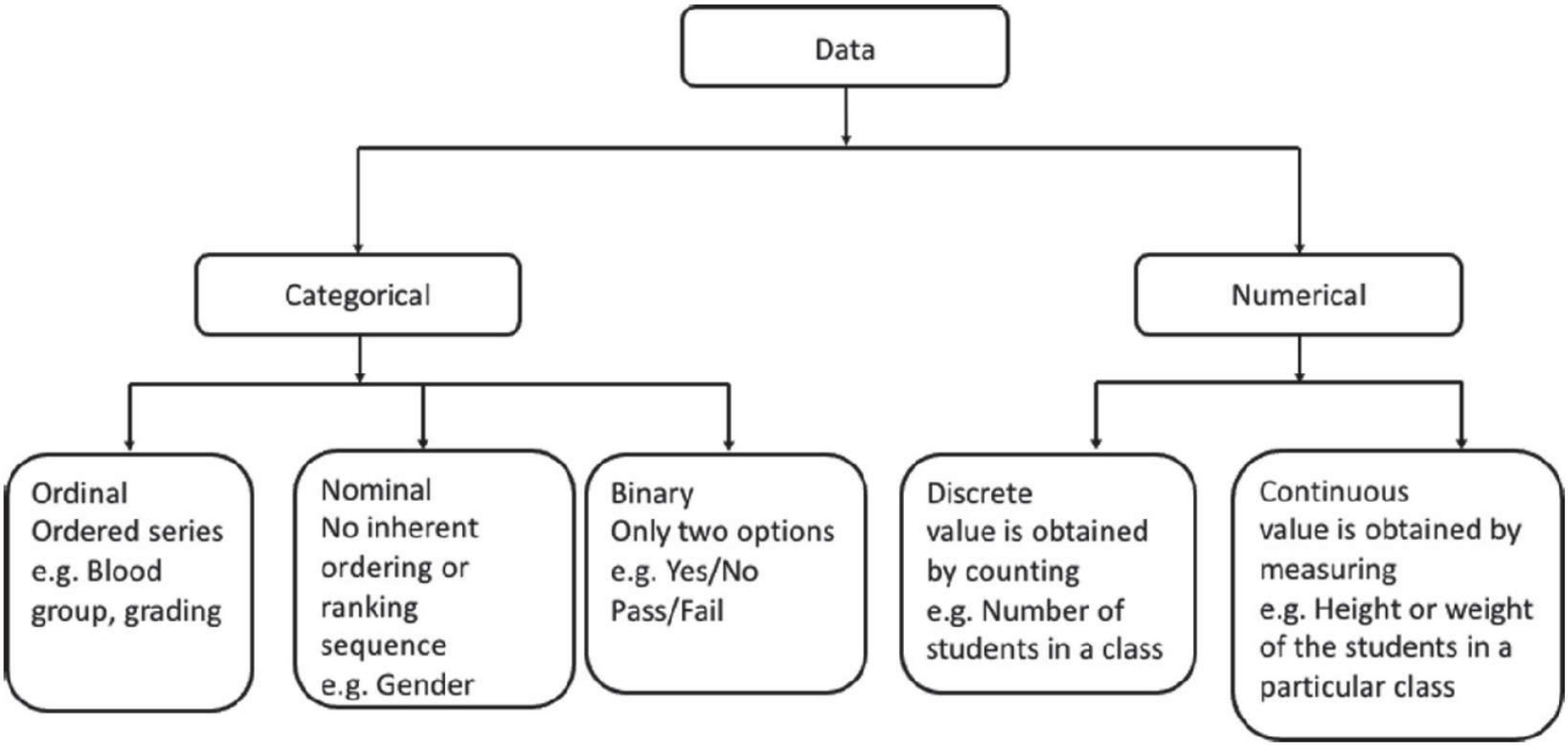
Variable: Measurements or characteristics of an item

Qualitative variable: Measurements or information about qualities

Quantitative variable: Measurements or information about quantities or numerical items

- **Numerical data**: Data that can be measured is the numerical data, e.g. the weather data like the temperature in degrees, students marks in examination etc.
  - Numerical data can be classified further into **discrete** and **continuous** values. The discrete data take fixed values and continuous data can take any value. For example, **days in a week** and the **price of a house**. The days in a week can have only seven values; however, the price of the house can have any value.

- **Categorical data**: The data that cannot be measured is categorical data; it represents based on some shared features; e.g. **gender, blood group**, etc.


Data
Categorical
Numerical
Ordinal
Ordered series
e.g. Blood group, grading
Nominal
No inherent ordering or ranking sequence
e.g. Gender
Binary
Only two options
e.g. Yes/No
Pass/Fail
Discrete
value is obtained by counting
e.g. Number of students in a class
Continuous
value is obtained by measuring
e.g. Height or weight of the students in a particular class